

Krish Patel

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EDUCATION

The University of Texas at Austin

Austin, TX

Bachelor of Science in Electrical and Computer Engineering GPA: 3.92/4.00

Aug. 2025 – May 2028

Coursework: Embedded Systems, Linear Systems + Signals, Circuit Theory, Software Design I/II, Digital Logic Design, Linear Algebra, Differential Equations, Discrete Mathematics, Multivariable Calculus, Organic Chemistry I/II

EXPERIENCE

Neural Engineering Intern

August 2026 – Present

Nurovo

Austin, TX

- Designed EEG preprocessing and feature engineering pipeline, extracting 130+ time-domain and frequency-band features from open-source neural datasets.
- Applied t-SNE dimensionality reduction to features and evaluated K-means clustering with elbow method to estimate number of distinct pain-related clusters.

ML Neuro-Classification Engineer

Oct. 2025 – Present

Longhorn Neurotech

Austin, TX

- Built end-to-end EMG signal processing and SVM classification pipeline, achieving 85% accuracy on hand-gesture classification.
- Processed 8-channel EMG data, including data cleaning, subject-based train/test splitting, and window segmentation.
- Performed feature selection using correlation analysis to remove redundant features and improve model efficiency.
- Implemented game engine control logic, graphical rendering, collision detection, and a bilingual user interface.
- Designed a custom PCB to interconnect the microcontroller, external controls, display, and audio circuitry.

Electrical Engineer

Oct 2025 – May 2026

EcoCAR

Austin, TX

- Interfaced NVIDIA Jetson Orin with Raspberry Pi 4 to support real-time environmental perception.
- Utilized Raspberry Pi 4 to manage user system control, including haptic feedback via vibration motor.
- Integrated edge inference outputs with backend server infrastructure to stream detection data to tablet and mobile app interfaces for real-time visualization.
- Implemented audio data acquisition pipeline for honk detection using CNN-based binary classification model.

PROJECTS

Family Style | C/C++, ARM32, KiCAD, GPIO, SPI

March 2026 – April 2026

- Developed real-time embedded video game in C/C++ on an MSPM0+ microcontroller, integrating GPIO controls, ADC joystick input, SPI-driven ST7735 LCD graphics, and DAC audio output.
- Wrote low-level peripheral drivers for switches, LEDs, UART/IR communication, and audio playback; used timer interrupts and SysTick handlers to coordinate input processing, timed events, and sound generation.
- Designed a custom PCB to interconnect microcontroller, external controls, display, and audio circuitry.

Robotathon | Sensor Fusion, Bluetooth, Servo, Autonomous Navigation

May 2018 – May 2020

- Integrated color, IR/distance, and line sensors into navigation logic to guide autonomous maze navigation.
- Implemented Bluetooth command handling for manual/autonomous mode switching and servo actuation of ball-launching mechanism, achieving 100% target accuracy in competition.

TECHNICAL SKILLS

Languages: Python, C/C++, Verilog, Java, ARM Assembly,

Tools: Git, VS Code, CCS, Claude Code, Codex, Jupyter, KiCAD, LTSpice, Oscilloscope, Logic Analyzer

Frameworks/Libraries: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib

Eligible to work in the United States without restrictions.